

EcoSphere ESG Management Platform

Master Product Blueprint (Version 1)

1. Vision

EcoSphere is an enterprise ESG management platform. Companies continue doing their normal daily work; EcoSphere records operational activities, calculates ESG metrics automatically, motivates employees through gamification, monitors governance and provides executive analytics.

2. Core Principle

Daily Work → Recorded in EcoSphere → ESG Calculation → Dashboard → Reports → Decisions

Final User Roles

- Admin (CEO + ESG Manager)
- Department Manager (one manager per department)
- Employee
- Auditor

3. User Journey

Admin: Configures company, departments, employees, emission factors, policies, rewards, badges and monitors ESG.

Department Manager: Adds operational data, creates CSR/challenges, approves proofs, monitors department, resolves compliance issues.

Employee: Participates in CSR/challenges, uploads proof, accepts policies, earns XP, claims rewards.

Auditor: Performs audits, raises compliance issues, verifies resolutions.

4. Modules

Operations (Mini ERP): Purchase, Electricity, Fleet, Manufacturing, Waste.

Environmental: Carbon calculations using emission factors.

Social: CSR, Challenges, Employee Participation.

Governance: Policies, Audits, Compliance Issues.

Gamification: XP, Coins, Badges, Rewards, Leaderboards.

Analytics: ESG dashboards, reports and exports.

5. Detailed Pages

Admin

- Executive Dashboard
- Departments
- Department Details
- Rewards & Badges
- Master Data
- Settings
- Reports

Department Manager

- Dashboard
- Operations
- CSR Activities
- Challenges

- Employees
- Policies
- Compliance
- Analytics

Employee

- Dashboard
- Challenges
- CSR
- Rewards & Badges
- Leaderboard
- Policies
- Notifications
- Profile

Auditor

- Dashboard
- Audits
- Compliance Issues
- History

6. Operations (Mini ERP)

Managers enter operational records only for their own department. Supported entries: Purchase, Electricity Consumption, Fuel Usage, Fleet Usage, Manufacturing Logs, Waste Generation. Each record supports attachments (image, PDF, video, bill). Department is auto-filled from the logged-in manager.

7. Environmental Flow

Manager records operation → System fetches emission factor → Carbon emission calculated automatically → Carbon Transaction created → Department Environmental Score updated → Organization ESG Score updated.

8. Social Flow

Manager creates CSR activity/challenge → Employees join → Upload proof → Manager approves → XP & Coins awarded → Badge rules checked → Leaderboards updated.

9. Governance Flow

Admin publishes policy → Employees receive notification → Employees accept policy → Auditor performs inspections → If violation found, compliance issue is created → Manager resolves → Auditor verifies → Governance score updates.

10. Gamification Engine

XP: earned through CSR, challenges and training.

Coins: redeemable currency.

Badges: automatically assigned when rules are satisfied.

Rewards: claimed using coins.

Leaderboards: Company, Department, Monthly, CSR.

11. Automatic Calculations

- Carbon = Quantity × Emission Factor
- Department ESG = Environmental + Social + Governance weighted score
- Company ESG = Average of department ESG scores
- Badge Assignment = Triggered after every XP update
- Leaderboard = Recalculated after XP changes

- Reward Eligibility = Coins $\geq$ Reward Cost
- Policy Completion % = Accepted Employees / Total Employees
- CSR Participation % = Participants / Eligible Employees
- Compliance Score = Closed Issues vs Total Issues

Notification Events

- New Policy Published
- Challenge Assigned
- Challenge Approved
- CSR Approved
- Badge Unlocked
- Reward Available
- Compliance Issue Assigned
- Issue Deadline Reminder
- Audit Completed

File Upload Support

- Images
- Videos
- PDF Bills
- Certificates
- Proof Documents

12. Future AI Layer

AI Chatbot for querying ESG data, AI executive report summaries, AI sustainability recommendations and predictive analytics. AI remains independent from the core platform.

13. Implementation Order

1. Freeze Product Design 2. Database Design 3. Backend APIs & Services 4. Authentication & RBAC 5. Frontend Dashboards 6. Notifications & File Uploads 7. Reports 8. AI Layer 9. Final Testing & Demo